FIRST-FREQUENCY ROTATION ON BINARY POINTED NECKLACES

ANONYMOUS

ABSTRACT. Let R be left rotation of a binary word and let $m_a(w) = |\{i : w_i = a\}|$. We study the finite map $T_n(w) = R^{m_{w_0}(w)}w$ on $\{0,1\}^n$. On a cyclic word of least period d and weight k, the pointed dynamics splits into $\gcd(k,d)$ disjoint $\pm k$ generator cycles. A constant component is periodic; every nonconstant component flows into its directed 10 edges. This gives the exact possible periods, the sharp maximum preperiod $n-2$, and exactly two deepest states for $n \geq 3$. Independently, we give the complete predecessor list of every target, the global 0/1/2 fibre distribution, and a primitive-block Möbius fixed census. Hamming-weight-controlled frozen rotations and the internal P166 cyclic phase architecture receive zero contribution credit. The retained status is `AMBER_INTERNAL_NEAR_P166 / HOLD_EXTERNAL`.

1. Literal map and subtraction boundary

For $n \geq 1$, write a word as $w = w_0 w_1 \cdots w_{n-1}$, with indices in $\mathbb{Z}/n\mathbb{Z}$, and set

$$R(w_0 w_1 \cdots w_{n-1}) = w_1 \cdots w_{n-1} w_0.$$

For $a \in \{0,1\}$, let $m_a(w) = |\{i : w_i = a\}|$. Our literal map is

$$T_n(w) = R^{m_{w_0}(w)}w, \qquad w \in \{0,1\}^n. \tag{1}$$

It preserves both the cyclic necklace and the weight $k = \mathrm{wt}(w) = m_1(w)$. On a fixed weight layer its two branches are

$$w_0 = 1: \quad T_n(w) = R^k w, \qquad w_0 = 0: \quad T_n(w) = R^{n-k}w = R^{-k}w. \tag{2}$$

Høyer and Špalek explicitly construct the quantum phase rotation $R_z(\varphi|x|)$ under the heading "rotation by Hamming weight" [2, Sec. 3.2]. Their object is not a cyclic coordinate shift. Nevertheless, we assign the Hamming-weight-controlled rotation idea and both frozen branches in (2) zero contribution credit. The adaptive first-symbol gluing is the only literal feature under analysis.

There is a closer internal boundary. P166 studies $S_n(x) = x + \mathrm{wt}(x)\mathbf{1}$ on $(\mathbb{Z}/n\mathbb{Z})^n$ and, on a diagonal-translation orbit, reduces it to $j \mapsto j + c_j$, where (c_j) is a weak composition of n. Our reduction below also has cyclic phase syntax, but its increments are the two values $\pm k$. The exact separation is summarized here.

axis	map (1)	internal P166 map
invariant orbit	coordinate-rotation necklace	diagonal alphabet-translation orbit
phase profile	$a_j \in \{k, -k\}$	$c_j \geq 0$, $\sum_j c_j = n$
forward engine	disjoint $\pm k$ generator cycles	cycle mass exhaustion
recurrence	possibly several nontrivial components	at most one nontrivial cycle per orbit
periods	$1, 2$, and proper divisors of n	every value $1, \ldots, n$
one-step fibres	two labelled rotations; size $0, 1, 2$	histogram-selected translates; unbounded in n

The common phase syntax, generic functional-graph language, the numerical clock value $n-2$, and indicator-style inverse notation all receive zero contribution credit. We retain only the constrained generator-component theorem and its consequences proved below. A direct owner of the adaptive literal map, a literal conjugacy into P166, or a proof that the component theorem is a formal consequence of P166 mass exhaustion triggers `KILL_INTERNAL_P166_PHASE_MAP`. Until that gate is resolved, the status remains

`AMBER_INTERNAL_NEAR_P166 / HOLD_EXTERNAL`.

For a point w of a finite self-map, its *preperiod* is the least $t \geq 0$ for which $T_n^t(w)$ is periodic. Its eventual exact period is called simply its period.

2. Pointed-necklace component theorem

Let $u \in \{0,1\}^n$ have least rotational period $d \mid n$. Its distinct pointed states are $R^j u$, $j \in \mathbb{Z}/d\mathbb{Z}$. We use the periodic indexing $u_{j+d} = u_j$ and put $k = \mathrm{wt}(u)$.

Lemma 2.1 (exact pointed reduction)**.** *On the rotation class of u, the identification $j \leftrightarrow R^j u$ conjugates T_n to*

$$\phi_u(j) = \begin{cases} j+k, & u_j = 1, \\ j-k, & u_j = 0, \end{cases} \qquad j \in \mathbb{Z}/d\mathbb{Z}. \tag{3}$$

Proof. The weight is constant on the rotation class. At $R^j u$, the first bit is u_j. Equations (2) therefore give increments k and $n-k$. Since $d \mid n$, the latter is $-k$ modulo d. □

Put

$$h = \gcd(k,d), \qquad L = d/h.$$

The undirected Cayley graph on $\mathbb{Z}/d\mathbb{Z}$ with generator k consists of h cycles of length L. For a component with representative r, order its vertices as $r, r+k, \ldots, r+(L-1)k$ and set $b_q = u_{r+qk}$, with $q \in \mathbb{Z}/L\mathbb{Z}$. Thus a bit 1 points forward and a bit 0 points backward in this generator order.

Theorem 2.2 (complete generator-component dynamics)**.** *Each generator component is classified as follows.*

(i) If $L = 1$, its vertex is fixed.
(ii) If $L = 2$, its two vertices form one directed two-cycle.
(iii) If $L \geq 3$ and (b_q) is constant, the component is one directed L-cycle.
(iv) If $L \geq 3$ and (b_q) is nonconstant, its recurrent components are exactly the cyclic occurrences $b_q b_{q+1} = 10$, one two-cycle per occurrence. Every other vertex is transient. More precisely, its preperiod is

$$\tau(q) = \begin{cases} \min\{t \geq 0 : (b_{q+t}, b_{q+t+1}) = (1,0)\}, & b_q = 1, \\ \min\{t \geq 0 : (b_{q-t-1}, b_{q-t}) = (1,0)\}, & b_q = 0. \end{cases} \tag{4}$$

Consequently the maximum preperiod in the component is one less than the longest cyclic constant run in (b_q).

Proof. The first two cases follow because $+1$ and -1 in generator coordinates are equal when $L \leq 2$. For $L \geq 3$, a constant word points coherently around the cycle and gives an L-cycle.

Now suppose the component word is nonconstant. Whenever $b_q b_{q+1} = 10$, the vertex q points to $q+1$ and $q+1$ points back to q, so this edge is a directed two-cycle. Starting from a 1, successive arrows move forward through its 1-run until the terminal 1 of such an edge is reached. Starting from a 0, they move backward through its 0-run until the initial 0 of such an edge is reached. This proves (4) and shows that every vertex reaches one of the listed edges.

No further directed cycle exists: a cycle that ever reverses direction uses an edge in both directions and is one of the listed two-cycles, while a cycle that never reverses would orient the whole component coherently, contrary to nonconstancy. On a constant run of length s, the farthest vertex is $s-1$ arrows from its boundary two-cycle. Taking the longest run proves the final statement. □

The theorem is pointwise, not merely a period count: factor the pointed necklace into its h generator cycles, read their constant runs, and obtain every tail and recurrent component without iterating (1). Several recurrent components can occur in one necklace. For example, the six rotations of 111000 split into three directed two-cycles.

3. Period inventory and sharp clock

Theorem 3.1 (all possible periods)**.** *For $n = 1$, the possible-period set is $\{1\}$. For every $n \geq 2$, it is*

$$\{1,2\} \;\cup\; \{\ell : \ell \mid n,\; 3 \leq \ell < n\}. \tag{5}$$

Proof. Theorem 2.2 shows that a period longer than two must be the length $L = d/\gcd(k,d)$ of a constant generator component. Hence $L \mid n$. If $L = n$, then $d = n$ and $\gcd(k,n) = 1$, so there is only one generator component. Its constancy would make the whole word constant, contradicting least period $d = n > 1$. Thus every long period is a proper divisor of n.

Constant words realize period one. For $n \geq 2$, the rotation class with a single 1 contains a directed two-cycle, realizing period two. It remains to realize a proper divisor $L \geq 3$. Write $n = gL$, where $g \geq 2$, and take the n-word of weight g whose 1-support is

$$\{0, 1, \ldots, g-2, g\} \subset \mathbb{Z}/n\mathbb{Z}. \tag{6}$$

The cyclic zero gap after position g is uniquely longest, so a rotation fixing the word must fix that gap and hence is trivial; the word has least period n. Under generator $k = g$, the component congruent to $g-1$ modulo g is all zero. It is therefore a directed cycle of length $n/g = L$ by Theorem 2.2. This realizes every period in (5). □

Let H_n be the maximum preperiod of T_n over $\{0,1\}^n$.

Theorem 3.2 (sharp clock and deepest census)**.** *One has*

$$H_1 = 0, \qquad H_n = n-2 \quad (n \geq 2). \tag{7}$$

For every $n \geq 3$, exactly two states have preperiod $n-2$, namely

$$010^{n-2} \quad \textit{and its bitwise complement.} \tag{8}$$

At the boundaries, both states are deepest for $n = 1$ and all four states are deepest for $n = 2$.

Proof. In a nonconstant generator component of length L, a constant run has length at most $L-1$. Theorem 2.2 bounds its preperiod by $L-2 \leq n-2$; constant components contain no transient vertex. This gives the upper bound for $n \geq 2$. For $n \geq 3$, the first word in (8) has weight one and, in generator order, a cyclic zero run of length $n-1$. Its displayed pointer is $n-2$ steps from the recurrent 10 edge. Complementation commutes with (1), so the second word is another witness.

Equality in the upper bound forces $L = n$ and a constant run of length $n-1$. Hence the word has exactly one minority bit. There are precisely two such rotation necklaces, one for each minority symbol, and in each there is exactly one pointer farthest from the recurrent 10 edge. These are the two words in (8). Directly, both length-one words are fixed, while at length two the constants are fixed and the two nonconstant words form a two-cycle. The boundary claims follow. □

The numerical value $n-2$ in (7) is not credited: P166 already has that sharp value. The retained statement is the generator-run proof together with the exact two-state last shell.

4. Every-target fibres and fixed census

For a target $y \in \{0,1\}^n$, subscripts such as y_{-k} are interpreted modulo n. Let $N_r(n,k)$ be the number of weight-k targets having exactly r one-step predecessors.

Theorem 4.1 (labelled inverse atlas)**.** *If y is constant, then $T_n^{-1}(y) = \{y\}$. If $0 < k = \mathrm{wt}(y) < n$, then its complete predecessor list is*

$$x_1 = R^{-k}y \qquad \textit{if and only if } y_{-k} = 1, \tag{9}$$

$$x_0 = R^{k}y \qquad \textit{if and only if } y_{k} = 0. \tag{10}$$

Here the subscripts label the first bit of the source, so present sources are distinct. In particular every fibre has size 0, 1, or 2.

For $0 < k < n$, if $n \mid 2k$, then

$$N_1(n,k) = \binom{n}{k}, \qquad N_0(n,k) = N_2(n,k) = 0. \tag{11}$$

Otherwise,

$$N_0(n,k) = N_2(n,k) = \binom{n-2}{k-1}, \qquad N_1(n,k) = \binom{n}{k} - 2\binom{n-2}{k-1}. \tag{12}$$

The two constant targets contribute two more to the global one-fibre count, and

$$|\operatorname{Im} T_n| = 2 + \sum_{k=1}^{n-1}\left[\binom{n}{k} - \mathbf{1}_{\{n\nmid 2k\}}\binom{n-2}{k-1}\right]. \tag{13}$$

Proof. A source beginning in 1 must use the first branch of (2); undoing it gives the sole candidate $R^{-k}y$, whose first bit is y_{-k}. Similarly, a source beginning in 0 uses rotation $n-k$, whose inverse is R^k; its first bit is y_k. This proves (9)–(10). The two candidates, when present, start with different bits.

The inspected positions $-k$ and k coincide precisely when $n \mid 2k$. Then exactly one of the complementary branch conditions holds, proving (11). Otherwise they are distinct. Fibre zero prescribes the ordered pair $(y_{-k}, y_k) = (0,1)$, and fibre two prescribes $(1,0)$. In either case the remaining $n-2$ positions contain $k-1$ ones, giving $\binom{n-2}{k-1}$. Subtraction gives the one-fibre count. Finally, subtracting the zero-fibre targets from each weight layer and adding the two constants proves (13). □

Fixed-density necklace and primitive-word enumeration are classical; see, for example, Heckenberger–Sawada [1] and Meštrović [3]. We use that background only to record a supporting fixed census.

Theorem 4.2 (Möbius fixed census). *Let μ be the Möbius function and define*

$$A(d,j) = \sum_{e \mid \gcd(d,j)} \mu(e) \binom{d/e}{j/e}. \tag{14}$$

Then the number of fixed points of T_n is

$$|\operatorname{Fix}(T_n)| = \sum_{d \mid n} \sum_{j=0}^{d} \mathbf{1}_{\{d \mid (n/d)j\}} A(d,j). \tag{15}$$

Proof. By Möbius inversion, $A(d,j)$ counts length-d binary linear words of least period d and weight j. Repeating such a primitive block n/d times gives a length-n word of least period d and total weight $k = (n/d)j$, and every length-n word arises uniquely in this way.

If the pointed first bit is 1, the word is fixed exactly when $d \mid k$. If it is 0, it is fixed exactly when $d \mid n-k$, which is equivalent because $d \mid n$. Thus the fixed condition is independent of the pointer and is exactly the indicator in (15). Summing the primitive blocks proves the formula. □

Primitive-word enumeration and Möbius inversion in (14)–(15) receive zero contribution credit.

5. Exact control and lifecycle

A paper-local, standard-library verifier starts from the literal update (1) and enumerates all 2^n words for every $1 \le n \le 18$. It checks carrier closure and weight preservation; every target in (9)–(10); the full fibre histogram; every pointed-necklace conjugacy; every component tail and recurrent cycle; the possible-period set; the sharp clock and deepest census; and (15). The frozen transcript reports 2,828,503 assertions and has SHA-256

`71720878a3498347661bad83838c3dbbc47c5c64c76ad7b70f6d5f02e7029190`.

Finite enumeration is a falsification and regression control, not a substitute for any all-parameter proof.

The retained results are precisely the $\pm k$ pointed-necklace component theorem, the period inventory, the sharp clock with deepest two, the every-target 0/1/2 fibres, and the Möbius fixed census. No external action is authorized. The terminal status is

$\boxed{\texttt{AMBER\_INTERNAL\_NEAR\_P166 / HOLD\_EXTERNAL}}$.

References

[1] I. Heckenberger and J. Sawada. A pascal-like bound for the number of necklaces with fixed density, 2018. URL https://arxiv.org/abs/1801.09516.

[2] Peter Høyer and Robert Špalek. Quantum fan-out is powerful. *Theory of Computing*, 1(5):81–103, 2005. doi: 10.4086/toc.2005.v001a005. URL https://theoryofcomputing.org/articles/v001a005/.

[3] Romeo Meštrović. Different classes of binary necklaces and a combinatorial method for their enumerations, 2018. URL https://arxiv.org/abs/1804.00992.